SIMATS ENGINEERING

ASSESSMENT TOOL 2: Problem Solving Assignment

COURSE NAME:	Digital Forensics and Cybercrime Investigation	**COURSE CODE:**	CSA61

COURSE OUTCOME COVERED

CO3: *Demonstrate digital evidence acquisition, preservation, validation, and forensic imaging using standard forensic procedures while maintaining evidence integrity.* (BL3)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Articulation (Level 4) Question Count: 3	**Duration: 60 Minutes**	**Total Marks: 30**

Code of Conduct

I ______________________(Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ______________________

Problem Solving Assignment Task

1. A laptop is seized with its RAM still active. Explain why memory should be acquired before disk imaging, referencing the concept of volatile vs non-volatile evidence.
2. Compare FAT, NTFS, and EXT file systems in terms of metadata storage and their impact on the forensic recoverability of deleted files.
3. Given a scenario with limited storage capacity at the seizure site, propose whether live or dead acquisition is more appropriate, and justify your choice.

Signature of the Faculty **Total Marks Awarded:**

Problem Solving Assignment Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Question 1

A laptop is seized with its RAM still active. Explain why memory should be acquired before disk imaging, referencing the concept of volatile vs. non-volatile evidence.

Answer

Digital forensic investigations aim to collect and preserve all available evidence without altering its integrity. When a computer is seized while it is still powered on, the investigator must first acquire the contents of **Random Access Memory (RAM)** before creating a forensic image of the storage device. This follows the **Order of Volatility**, which states that evidence that can disappear quickly must be collected before evidence that remains permanently stored.

Volatile evidence is information that exists only while the system has power. Once the computer is shut down or restarted, this data is permanently lost. Examples include:

- Running processes and services
- Active network connections
- Logged-in user sessions
- Encryption keys and passwords
- Unsaved documents
- Malware executing only in memory
- Clipboard contents and temporary data

Non-volatile evidence is stored on permanent storage devices such as hard disks and SSDs. This data remains intact even after the computer is powered off. Examples include:

- Files and folders
- Installed applications
- Operating system files
- Email databases
- Browser history
- Event logs

If investigators immediately shut down the laptop and begin disk imaging, all valuable volatile evidence stored in RAM will disappear forever. Therefore, a **live memory acquisition** is performed first using trusted forensic tools such as **Belkasoft RAM Capturer, FTK Imager, Magnet RAM Capture, or DumpIt**. After capturing the memory, investigators create a **bit-by-bit forensic image** of the storage device using a write blocker to prevent any modification to the original evidence.

To ensure evidence integrity, cryptographic hash values such as **MD5** or **SHA-256** are generated before and after imaging. Matching hash values confirm that the forensic image is an exact copy of the original data.

Conclusion

Memory acquisition must always precede disk imaging because volatile evidence disappears immediately when power is lost. Collecting RAM first ensures that investigators preserve all available digital evidence while maintaining forensic integrity and legal admissibility.

Question 2

Compare FAT, NTFS, and EXT file systems in terms of metadata storage and their impact on the forensic recoverability of deleted files.

Answer

File systems organize data on storage devices and maintain metadata that is essential during forensic investigations. The amount of metadata stored by each file system directly affects the ability of investigators to recover deleted files and reconstruct user activities.

Feature	FAT (File Allocation Table)	NTFS (New Technology File System)	EXT (Extended File System)
Metadata Storage	Limited metadata	Extensive metadata stored in the Master File Table (MFT)	Metadata stored in inode structures
File Permissions	Not Supported	Supported	Supported
Journaling	No	Yes	Yes (EXT3/EXT4)
Security Features	Basic	Encryption, compression, access control	Linux permissions and ownership
Deleted File Recovery	High if not overwritten	Very good due to MFT records	Moderate to difficult depending on EXT version
Common Usage	USB drives, memory cards	Windows operating systems	Linux operating systems

FAT (File Allocation Table)

FAT stores only basic metadata such as filename, creation date, and file allocation information. When a file is deleted, the file entry is marked as deleted, but the actual data remains on the disk until it is overwritten. Because of its simple structure, deleted files can often be recovered successfully using forensic recovery tools.

NTFS (New Technology File System)

NTFS stores detailed metadata in the **Master File Table (MFT)**. Each file has an MFT record containing timestamps, permissions, ownership information, file attributes, and allocation details. Even after deletion, many MFT records remain available, making NTFS one of the best file systems for forensic investigations. It also supports journaling, allowing investigators to reconstruct system activities.

EXT (Extended File System)

EXT file systems used in Linux store metadata inside **inode structures**. Each inode records ownership, permissions, timestamps, and disk block locations. Modern versions such as EXT3 and EXT4 support journaling, which improves system reliability but may reduce the likelihood of recovering deleted files because metadata can be updated or removed quickly after deletion. Investigators generally require specialized Linux forensic tools to recover deleted data.

Comparison

- FAT provides simple metadata and is easier for deleted file recovery.
- NTFS stores the richest metadata and offers excellent forensic analysis capabilities.
- EXT provides strong security and journaling but requires advanced forensic techniques for deleted file recovery.

Conclusion

Among the three file systems, **NTFS** provides the most comprehensive forensic evidence because of its Master File Table and detailed metadata. FAT offers easier deleted-file recovery, while EXT provides strong security features but requires specialized forensic analysis.

Question 3

Given a scenario with limited storage capacity at the seizure site, propose whether live or dead acquisition is more appropriate, and justify your choice.

Answer

Digital evidence acquisition must balance evidence preservation, storage availability, and forensic soundness. When storage capacity at the seizure site is limited, investigators must decide whether **live**

acquisition or **dead acquisition** is the better approach.

Live Acquisition

Live acquisition is performed while the computer is still running.

Advantages

- Captures volatile evidence stored in RAM.
- Recovers encryption keys, passwords, running processes, and active network connections.
- Useful for encrypted systems and live malware analysis.

Disadvantages

- Changes may occur on the system during acquisition.
- Requires additional storage for memory dumps.
- Greater risk of altering evidence.

Dead Acquisition

Dead acquisition is performed after the computer has been safely powered down.

Advantages

- Produces a complete bit-by-bit forensic image.
- Preserves evidence integrity.
- Easier to maintain the chain of custody.
- Requires less temporary storage than capturing both RAM and disk simultaneously.
- Preferred in most standard forensic investigations.

Disadvantages

- Volatile evidence stored in RAM is permanently lost.

Recommended Approach

When storage capacity at the seizure site is limited, **dead acquisition is generally the preferred method**, provided that the system does not contain critical volatile evidence. Investigators can remove the storage device, create a forensic image using write blockers, and perform detailed forensic analysis later in a controlled laboratory environment.

However, if the computer is powered on and contains encrypted volumes, active malware, or important unsaved information, investigators should first perform a **targeted live acquisition of RAM**, followed by dead acquisition of the storage device.

Conclusion

Under limited storage conditions, **dead acquisition is the most appropriate choice** because it preserves evidence integrity, simplifies forensic imaging, and minimizes storage requirements. Live acquisition should only be used when volatile evidence is essential to the investigation.